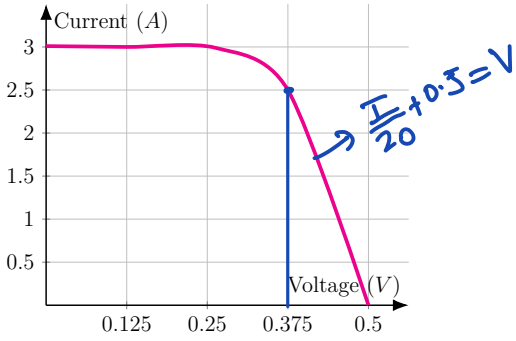
EE2015: Electric Circuits and Networks

Tutorial 2

(August 13, 2025)

1. The current-voltage characteristic of a silicon solar cell exposed to direct sunlight is shown on the right. It is obtained by placing different-sized resistors across the two terminals of the device and measuring the resulting currents and voltages.

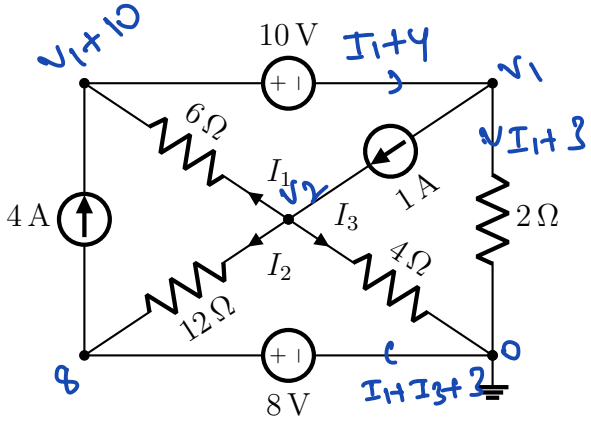


   (a) What is the value of the short-circuit current? 3A

   (b) What is value of the voltage at open circuit? 12V

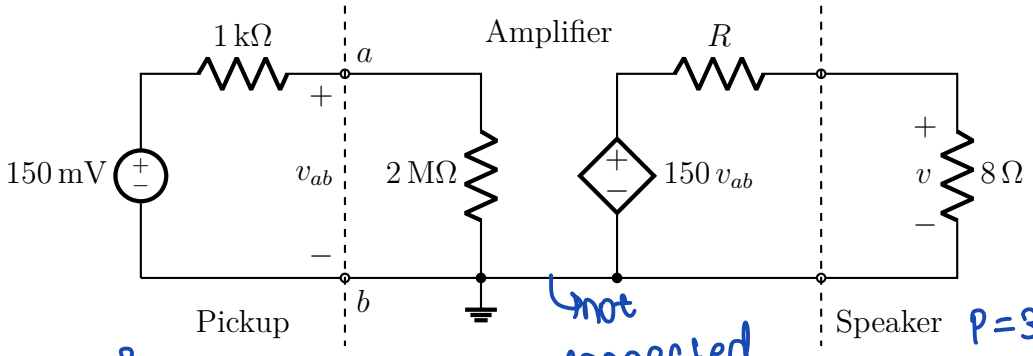
   (c) Estimate the maximum power that can be obtained from the device? 0.9375 W

2.



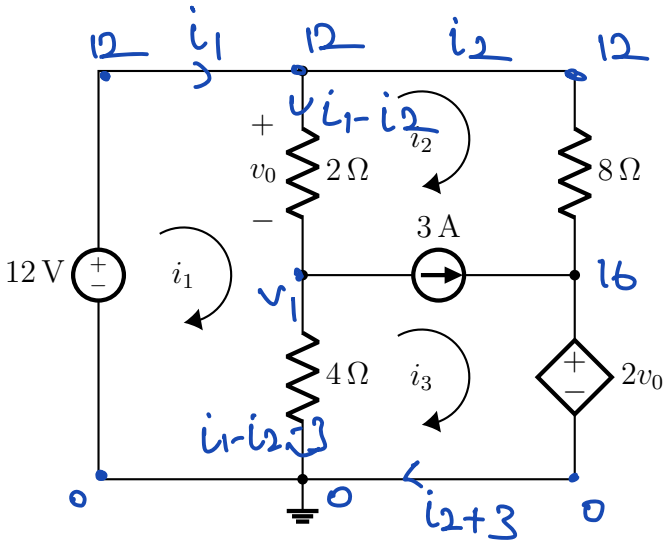
Consider the circuit shown on the left. What are the values of I_1, I_2, and I_3?

3. The equivalent circuit of a phonograph pickup, stereo amplifier, and speaker connected in series is as show below. Determine the resistance R so that the voltage v across the speaker is 16 V. What is the power delivered to the speaker?



4.

For the circuit shown on the right, find i_1, i_2, and i_3.



$-12+2(i_1-i_2)+4(i_1-i_2-3)=0$

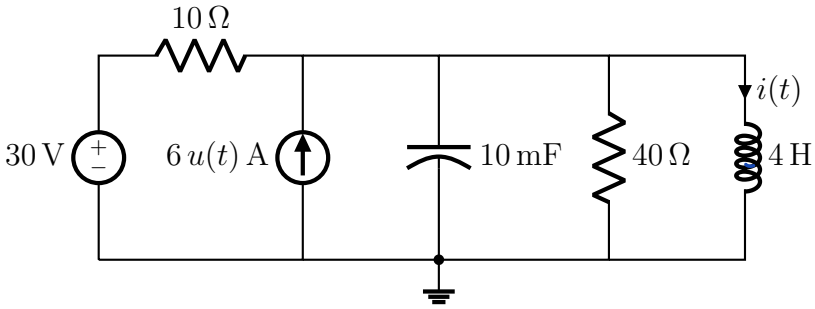
$6(i_1-i_2)=24$

$i_1-i_2=4 \Rightarrow v_0=8$

$i_2=-4/8=-1/2 \Rightarrow i_1=7/2 \quad i_3=5/2$

5.

Consider the circuit shown on the left. Find $i(t)$ for $t \geq 0$.



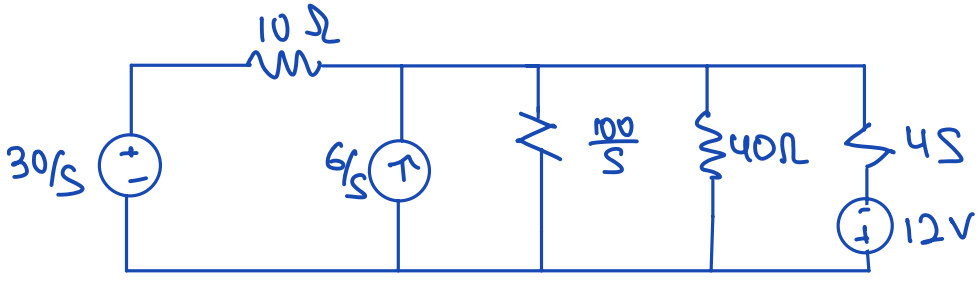
$t<0$ $6u(t)=0$

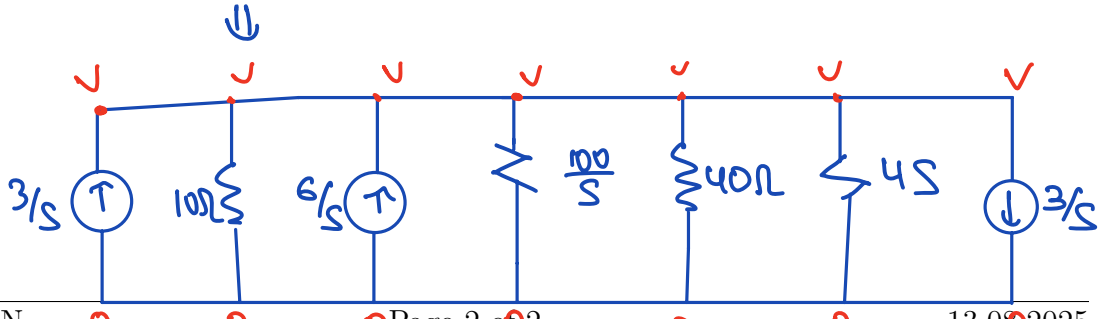
capacitor → open

inductor → shorted

$\Rightarrow i = 3A$ (at $t=0^-$)

$\Rightarrow i(0)=3A$





$3/s+6/s = V/10 + V/(100/s) + V/40 + V/4s + 3/s$

$V\left[\frac{1}{10}+\frac{s}{100}+\frac{1}{40}+\frac{1}{4s}\right]=\frac{6}{s} \Rightarrow V=\frac{600}{s^2+12.5s+25}$

$I(s)=\frac{V(s)}{4s}+\frac{3}{s}=\frac{150}{s(s^2+12.5s+25)}+\frac{3}{s}=\frac{9}{s}-\frac{8}{s+2.5}+\frac{2}{s+10}$

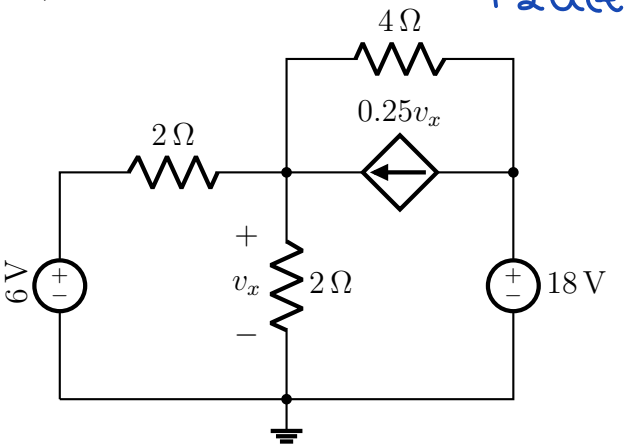
$I(t)=9u(t)-8u(t)e^{-2.5t}+2u(t)e^{-10t}$

EE2015: Electric Circuits and Networks

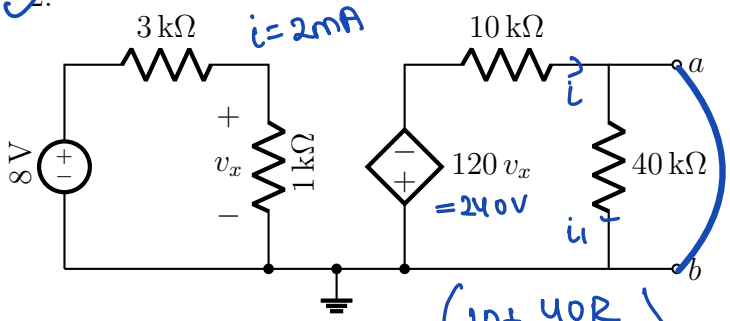
Tutorial 3

(August 22, 2025)

1. For the circuit shown on the right, find v_x by taking advantage of source transformation.



2.



(a) For the circuit shown on the left, what resistor connected across the output terminals a-b will absorb maximum power from the circuit? =8kΩ

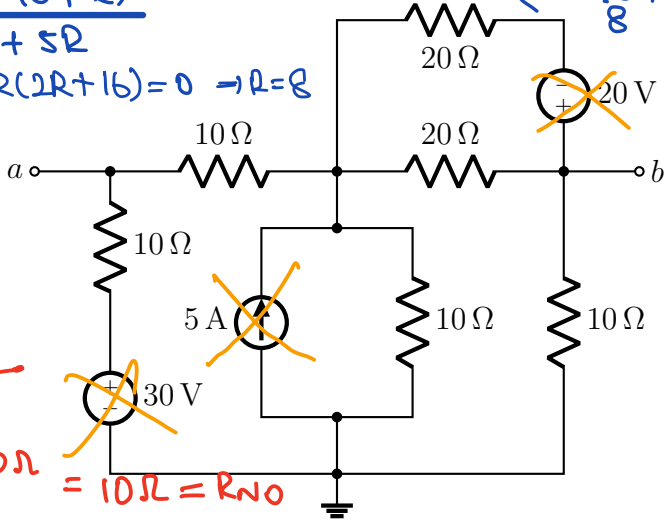
(b) What is the maximum value of power absorbed by the load? 1.152W

$i=\frac{24}{10+\frac{40R}{40+R}}=\frac{24(40+R)}{40+5R}$

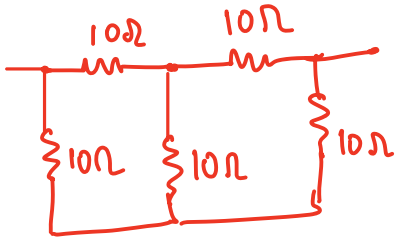
$(24-i)10=\frac{10}{2}\left(\frac{24\times 4R}{40+8R}\right)$

$\frac{192^2 R}{(R+8)^2}\to max \Rightarrow \frac{R}{(R+8)^2}\to max \Rightarrow (R+8)^2-R(2R+16)=0 \Rightarrow R=8$

3. For the circuit shown on the right, find the Norton equivalent as seen across terminals a-b.



R_{th}:



$=10\Omega=R_{NO}$

$-3+10^3 I_o + 2v_x = 0$

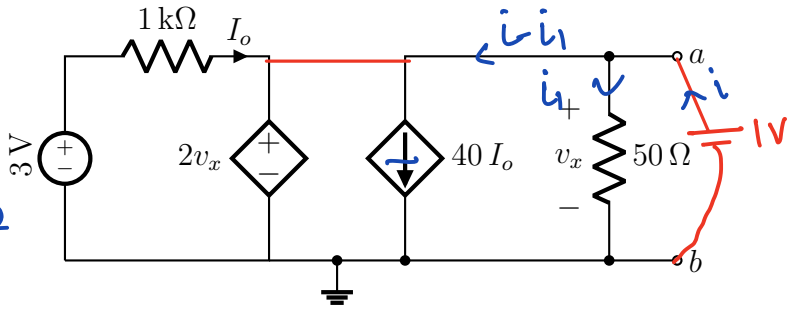
$\Rightarrow -3 - v_x/2 + 2v_x = 0$

$\Rightarrow 3v_x/2 = 3 \Rightarrow v_x = 2V$

$v_x/50 = 40 I_o \Rightarrow v_x = 2\times10^3 I_o$

$I_o = 1mA$

4. For the circuit shown on the right, find the Thevenin equivalent as seen across terminals a-b.



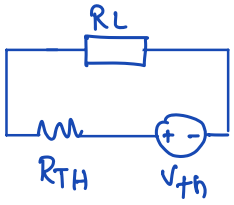
$1 = 50 i_1 \Rightarrow i_1 = 0.02$

$i - i_1 = 0.04 \Rightarrow i = 0.06$

5. An uknown ciruit has output terminals a-b across which a variable resistor R_L is connected. The load current I_L and load voltage V_L are tabulated below for three different values of R_L.

$R/(R+10)^2$ $\quad (R+10)^2 - 2R(R+10) \Rightarrow R = 10$

R_L (ohms)	V_L (volts)	I_L (amps)
2	6	3
8	16	2
14	21	1.5



$i_L = \frac{V_{th}}{R_{TH}+R_L}$

$3 = \frac{V_{th}}{R_{th}+2}$

$1.5 = \frac{V_{th}}{R_{th}+14}$

$R_{th} = 10$

$V_{th} = 36$

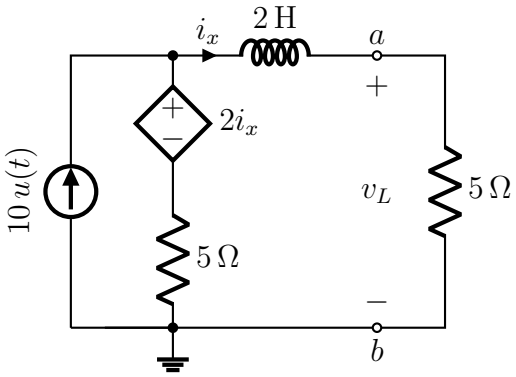
$\left(\frac{36}{10+R}\right)^2 R$

(a) Find I_L when $R_L = 4\,\Omega$. $36/14 = 18/7\ A$

$2 = \frac{R_{th}+14}{R_{th}+2} \Rightarrow \frac{12}{R_{th}+2} = 1$

(b) What is the maximum power that this unknown circuit is capable of supplying to the load ? $1.8^2 \times 10 = 32.4\,W$

6. In the circuit shown on the right, the 5 Ω load resistance is connected across a-b.



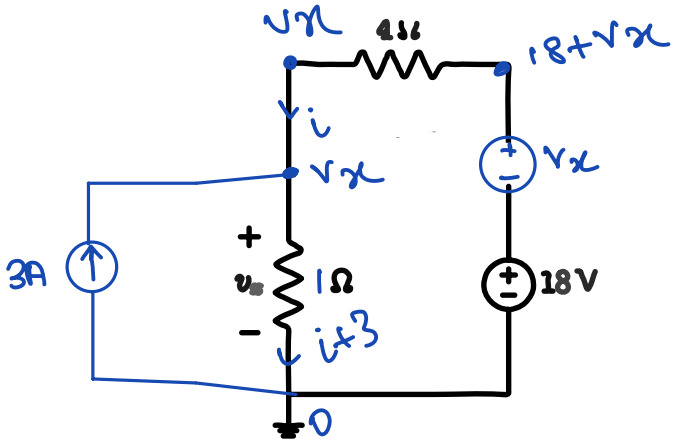
(a) Find the Thevenin voltage $V_{Th}(s)$. $5/s$

(b) Find $Z_{Th}(s)$. $2s+3$

(c) Find $v_L(0^+)$ and $v_L(\infty)$. 0

(d) Find $v_L(t)$ for $t \geq 0$.

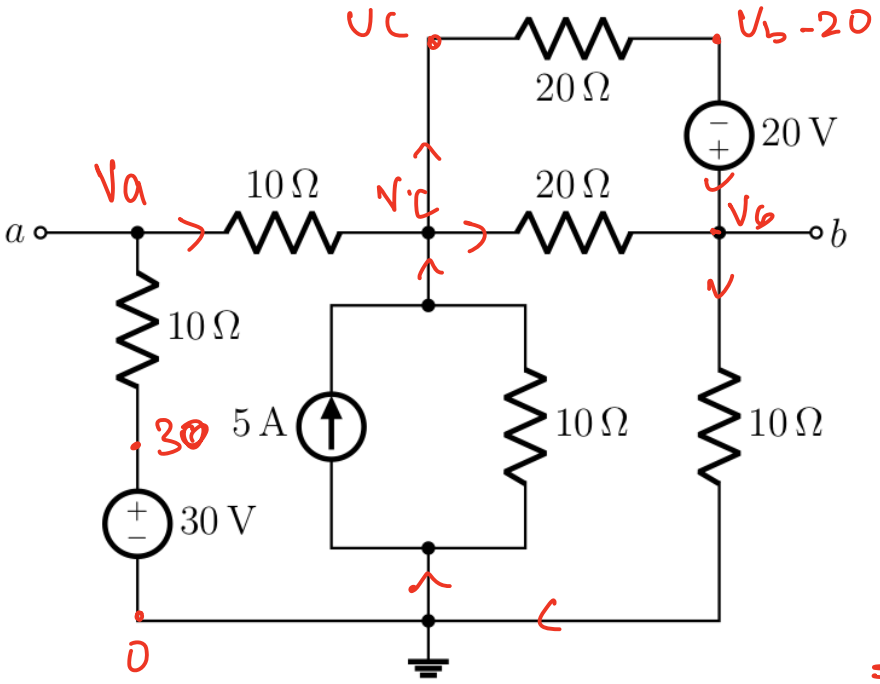
①



$$i = \frac{(18+V_x) - V_x}{4} = 4.5$$

$$\Rightarrow V_x = (i+3)\,1 = 7.5V$$

②



$$\frac{30-V_a}{10} = \frac{V_a-V_c}{10}$$

$$\Rightarrow 30 = 2V_a - V_c$$

$$\frac{V_c-V_b}{20} + \frac{V_c-V_b+20}{20} = \frac{V_b}{10}$$

$$\Rightarrow 2V_b - V_c = 10$$

$$\frac{V_a-V_c}{10} - \frac{V_c}{10} = \frac{V_b}{10} \Rightarrow V_a = V_b + 2V_c$$

$$\frac{30+V_c}{2} = \frac{V_c+10}{2} + 2V_c \Rightarrow V_c = 5$$

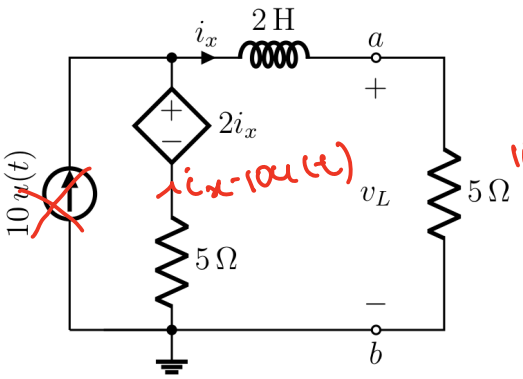
$$\Rightarrow V_b = 7.5$$

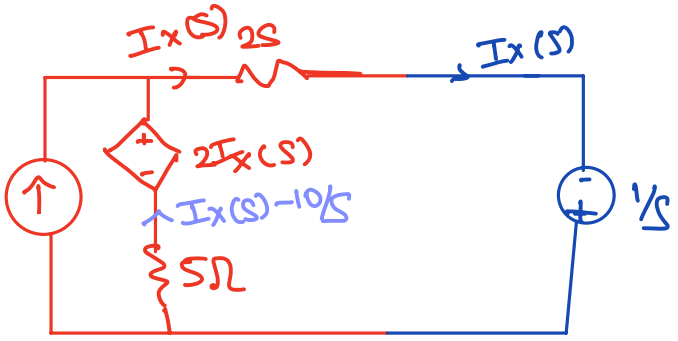
$$V_a = 17.5$$

$$V_a - V_b = 10$$

$$\Rightarrow I_{N0} = 10/10 = 1A$$

⑥



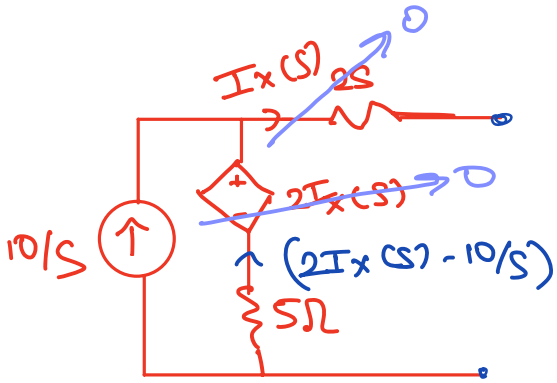


$$2I_x(S) + \frac{1}{S} = (2S+5)I_x(S)$$

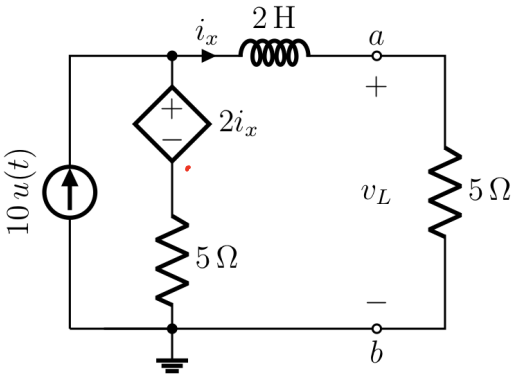
$$\Rightarrow (2S+3)I_x(S) = \frac{1}{S}$$

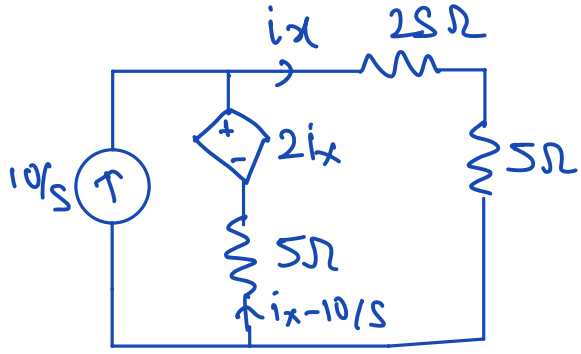
$$\downarrow$$

$$R_{th}$$



$$V_{th} = \frac{50}{S}$$





$$2i_x = (2S+5)\,i_x + (i_x - 10/S)5$$

$$\frac{25}{S} = (S+4)\,i_x \rightarrow \frac{25}{S(S+4)} = i_x$$

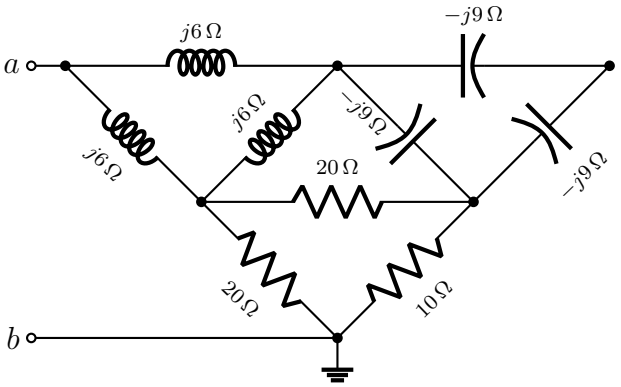
$$\Rightarrow \frac{125}{4}\left\{\frac{1}{S} - \frac{1}{S+4}\right\} = V_L(S)$$

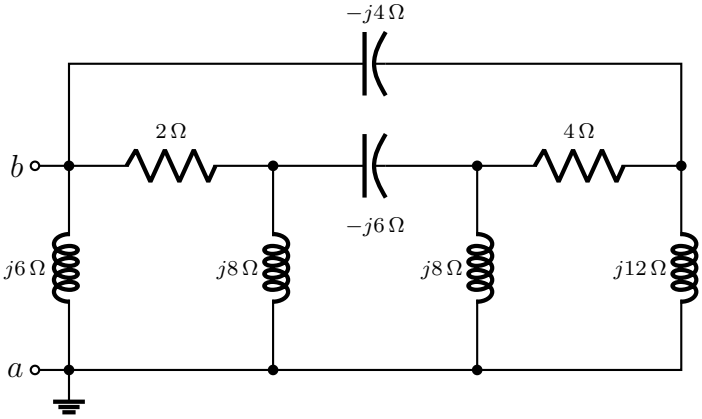
EE2015: Electric Circuits and Networks

Tutorial 4

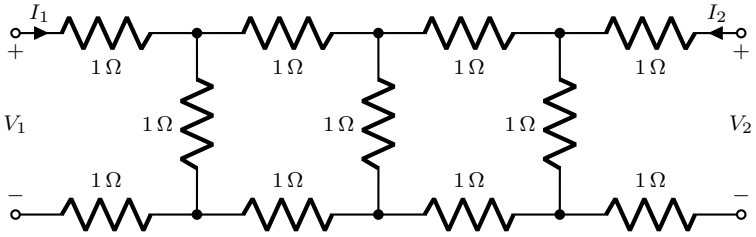
(September 12, 2025)

1. Calculate the impedance $\mathbf{Z}_{ab}$ of the one-port networks shown below.

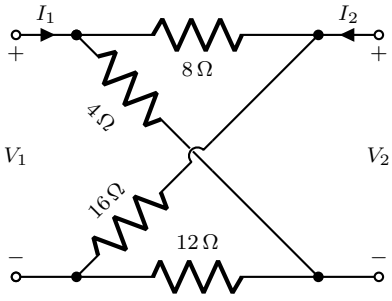




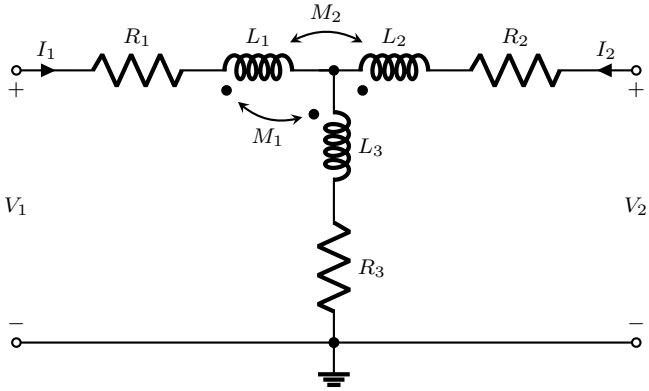
2. For the two-port network shown on the right find the impedance and admittance parameters.



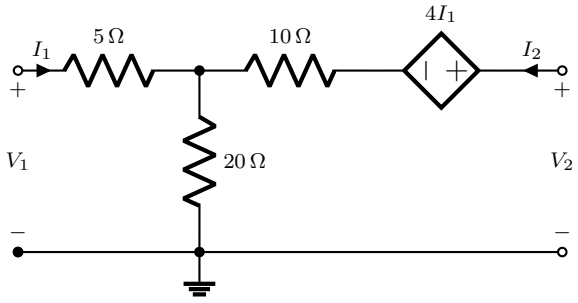
3. Determine the z- and y-parameters of the two-port network shown on the left.

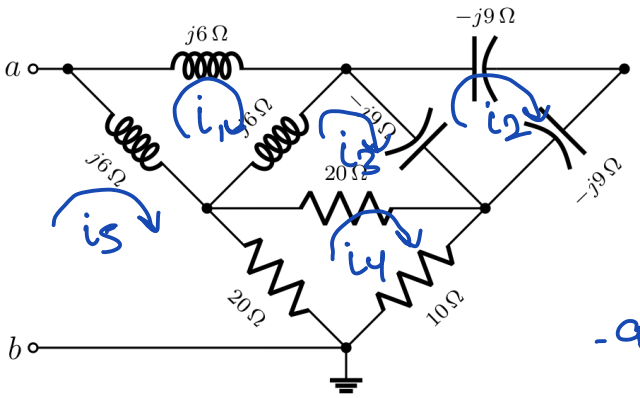


4. Find the entries of $\mathbf{Z}(s)$ of the two-port network shown on the right.



5. Find the z-parameters for the two-port network shown on the right.





$6j\,i_1 + 6j(i_1 - i_3) + 6j(i_1 - i_5) = 0$

$-9j\,i_2 - 9j\,i_2 - 9j(i_2 - i_3) = 0$

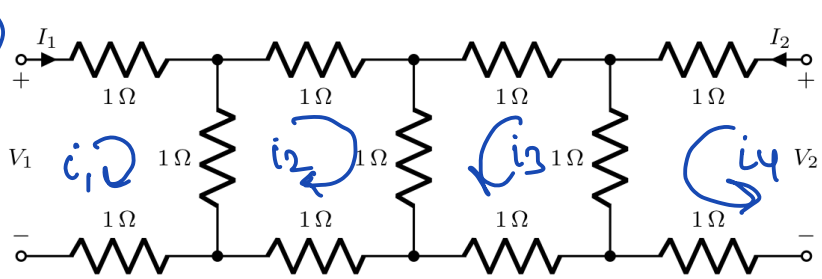
$10i_4 + 20(i_4 - i_5) + 20(i_4 - i_2) = 0$

$-9j(i_3 - i_2) + 20(i_3 - i_4) + 6j(i_3 - i_1) = 0$

$V_1 = 6j(i_5 - i_1) + 20(i_5 - i_3)$

$(-6j)$	0	(-20)	0	$(6j+20)$	V_1
$(18j)$	0	$(-6j)$	0	$(-6j)$	0
0	$(-27j)$	$(9j)$	0	0	0
0	0	(-20)	50	-20	0
$(-6j)$	$(9j)$	$(-3j+20)$	-20	0	0

(2)



$V_1 - I_1 - I_1 + I_2 - I_1 = 0$

$I_2 - I_1 + I_2 + I_2 + I_3 + I_2 = 0$

$I_3 + I_2 + I_3 + I_3 + I_3 - I_4 = 0$

$V_2 - I_4 - I_4 + I_3 - I_4 = 0$

$3I_1 - I_2 = V_1$ $\quad 4I_2 + I_3 - I_1 = 0$

$3I_4 - I_3 = V_2$ $\quad 4I_3 + I_2 - I_4 = 0$

$$4\{3I_1 - V_1\} + \{3I_4 - V_2\} - I_1 = 0$$

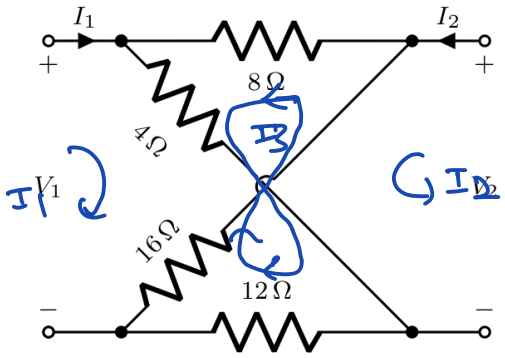
$$4\{3I_4 - V_2\} + \{3I_1 - V_1\} - I_4 = 0$$

$$\Rightarrow 11I_1 + 3I_4 = 4V_1 + V_2$$

$$\Rightarrow 3I_1 + 11I_4 = V_1 + 4V_2$$

$$\Rightarrow I_1 = \frac{41V_1 - V_2}{112} \qquad I_2 = \frac{41V_2 - V_1}{112}$$

$$\Rightarrow Y = \begin{bmatrix} \frac{41}{112} & \frac{-1}{112} \\ \frac{-1}{112} & \frac{41}{112} \end{bmatrix} \qquad Z = Y^{-1} = \frac{112}{15}\begin{bmatrix} \frac{41}{112} & \frac{1}{112} \\ \frac{1}{112} & \frac{41}{112} \end{bmatrix} = \begin{bmatrix} \frac{41}{15} & \frac{1}{15} \\ \frac{1}{15} & \frac{41}{15} \end{bmatrix}$$



=

$$40I_3 + 4I_1 - 16I_2 =$$

$$V_1 = 2.4I_1 + 6.4I_2$$

$$V_2 = 28I_1 + 5.8I_2$$